DC SYSTEM SIZE: 23 x 450 = 10.350KW DC
AC SYSTEM SIZE: 01 x 11500 = 11.500KW AC
ESS AC CAPACITY: 27 KWH
(23) SIRIUS SOLAR ELNSM54M-HC-N 450W PV MODULES
WITH (09) TESLA: RSD MCI-2
(01) TESLA POWERWALL 3 (1707000-XX-Y)
(01) TESLA POWERWALL 3 EXPANSION (1807000-xx-y)

NOTE: FMC MAY BE USED IN INDOOR APPLICATIONS WHERE PERMITTED BY NEC ART. 348.3

NOTE: CONDUIT TO BE UL LISTED FOR WET LOCATIONS AND UV PROTECTED (EX. - EMT, IMC 80 PVC OR RMC)

INSTALLER / ELECTRICIAN NOTE:
EC IS TO MEASURE VOLTAGE BEFORE STARTING WORK.
IF RESULT IS ANY OTHER VOLTAGE MEASURED THAN 120/240V IS OBSERVED, DO NOT PROCEED. CONTACT ENGINEER.

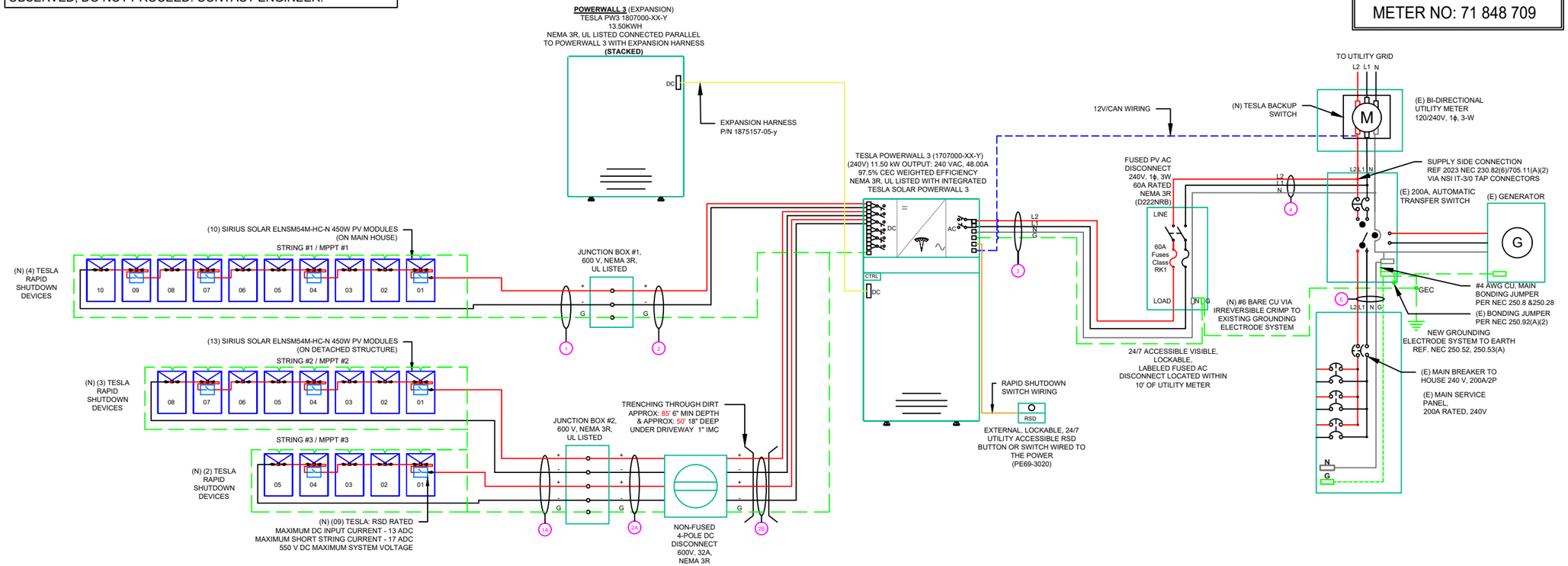
INTERCONNECTION NOTES:
1. INTERCONNECTION SIZING, LIMITATIONS AND COMPLIANCE DETERMINED IN ACCORDANCE WITH [NEC].
2. GROUND FAULT PROTECTION IN ACCORDANCE WITH [NEC].

DISCONNECT NOTES:
1. DISCONNECTING SWITCHES SHALL BE WIRED SUCH THAT WHEN THE SWITCH IS OPENED THE CONDUCTORS REMAINING LIVE ARE CONNECTED TO THE TERMINALS MARKED “LINE SIDE” (TYPICALLY THE UPPER TERMINALS)
2. AC DISCONNECT MUST BE ACCESSIBLE TO QUALIFIED UTILITY PERSONNEL, BE LOCKABLE, AND BE A VISIBLE-BREAK SWITCH
3. DISCONNECT MEANS AND THEIR LOCATION SHALL BE IN ACCORDANCE WITH [NEC 225.31] AND [NEC 225.32].

RACKING NOTE:
1. BOND EVERY OTHER RAIL WITH #6 BARE COPPER

GROUNDING & GENERAL NOTES:
1. **GROUNDING ELECTRODES AND GROUNDING ELECTRODE CONDUCTORS.** ADDITIONAL GROUNDING ELECTRODES SHALL BE PERMITTED TO BE INSTALLED IN ACCORDANCE WITH 250.52 AND 250.54.GROUNDING ELECTRODES SHALL BE PERMITTED TO BE CONNECTED DIRECTLY TO THE PV MODULE FRAME(S) OR SUPPORT STRUCTURE PER [NEC 690.47(B)]
2. PV INVERTER IS UNGROUNDED, TRANSFORMER-LESS TYPE.
3. DC GEC AND AC EGC TO REMAIN UNSPLICED, OR SPLICED TO EXISTING ELECTRODE
4. ANY EXISTING WIRING INVOLVED WITH PV SYSTEM CONNECTION THAT IS FOUND TO BE INADEQUATE PER CODE SHALL BE CORRECTED PRIOR TO FINAL INSPECTION.
5. JUNCTION BOX QUANTITIES, AND PLACEMENT SUBJECT TO CHANGE IN THE FIELD - JUNCTION BOX DEPICTED ON ELECTRICAL DIAGRAM REPRESENT WIRE TYPE TRANSITIONS.
6. AC DISCONNECT NOTED IN EQUIPMENT SCHEDULE OPTIONAL IF OTHER AC DISCONNECTING MEANS IS LOCATED WITHIN 10' OF SERVICE DISCONNECT.
7. RACEWAYS AND CABLES EXPOSED TO SUNLIGHT ON ROOFTOPS SHOULD BE INSTALLED MORE THAN 7/8" ABOVE THE ROOF USING CONDUIT SUPPORTS.

METER NO: 71 848 709



	QTY	CONDUCTOR INFORMATION		CONDUIT TYPE	CONDUIT SIZE
1	(2)	CU#10AWG -	PV WIRE/USE-2	N/A	N/A
	(1)	CU #6AWG -	BARE COPPER IN FREE AIR		
1A	(4)	CU#10AWG -	PV WIRE/USE-2	N/A	N/A
	(1)	CU #6AWG -	BARE COPPER IN FREE AIR		
2	(2)	CU#10AWG -	THWN-2 +VE & -VE	IMC OR RMC ONLY ON EXTERIOR, EMT IN ATTIC	3/4"
	(1)	CU #10AWG -	CU,THWN-2 GND		
2A	(4)	CU#10AWG -	THWN-2 +VE & -VE	IMC OR RMC ONLY ON EXTERIOR, EMT IN ATTIC	3/4"
	(1)	CU #10AWG -	CU,THWN-2 GND		
2B	(4)	CU#10AWG -	THWN-2 +VE & -VE	IMC (6" MIN DEPTH BURIAL)	1"
	(1)	CU #10AWG -	CU,THWN-2 GND		
3	(3)	CU#6AWG -	THWN-2 L1,L2,N	IMC ON EXTERIOR	3/4"
	(1)	CU #10AWG -	CU,THWN-2 GND		
4	(2)	CU#6AWG -	THWN-2 L1& L2	IMC ON EXTERIOR	3/4"
	(1)	CU#6AWG -	THWN-2 N		

NOTE : "CONDUIT SIZE IS MINIMUM REQUIRED PER NEC300.17. CONTRACTOR MAY UPSIZE AS NEEDED".

1 ELECTRICAL LINE DIAGRAM
PV-5 SCALE: NTS

TRON SOLAR

TRON SOLAR LLC.
CONTRACTOR ADDRESS:
950 CORPORATE WOODS PARKWAY,
VERNON HILLS, IL 60061
LICENSE #. TGC110144

REVISIONS		
DESCRIPTION	DATE	REV
INITIAL	07/20/2026	
REVISION	07/28/2026	A
REVISION	10/12/2026	B

PROJECT NAME & ADDRESS
SYLVIA MOYES RESIDENCE
308 S 3RD ST,
SAINT FRANCISVILLE, IL 62460
(38.589233, -87.643491)

DRAWN BY
ESR

SHEET NAME
ELECTRICAL LINE DIAGRAM

SHEET SIZE
ANSI B
11" X 17"

SHEET NUMBER
PV-5